

SAI AKILESH VENIGALLA

San Francisco, CA 94107 | 4+ Years of SWE Experience

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Experience

Software Engineer Intern

Jan 2026 - Present

Tesla | *Building mission-critical products and features for the RoboTaxi and Supercharging teams.*

- **Accelerated Robotaxi payments processing by 85%** by architecting a scalable **Kafka** pipeline and **Go** based DLQ for robust, asynchronous event processing and fault-tolerant retries.
- Achieved **90% faster ride data retrieval** via a high-throughput **MySQL** to **Elasticsearch** sync pipeline.
- Maximized Supercharger utility via 100% visibility of live wait-time forecasts to **Google Maps** via **gRPC** and **OCPI**.
- Scaled referral device fingerprinting to **3.5M+ DAU** using **Redis**, idempotency and a secure Apple Pay pre-auth engine.
- Leveraged **H3 geospatial indexing** to optimize Robotaxi surge pricing for a 400+ vehicle fleet, adjusting rates in real-time based on location, fleet congestion, and demand patterns.

Network Engineer

Jan 2025 - Present

NYU IT High Speed Research Network Lab | *Container Orchestration, Network Design and Research.*

- **Reduced environment setup time by 96%** (1hr to under 2mins) and cut **resource consumption by 50%** by containerizing a Kea DHCP, FastAPI, and PostgreSQL network management solution with Docker and Docker Compose.
- Eliminated 100% of licensing costs by developing Rest API interface for **Kea DHCP** service on **Linux/Unix Systems**.
- Strengthened network security by architecting a Syslog monitoring pipeline to **detect SSH logins**, triggering real-time alerts via Matrix/Element that provided 100% visibility across distributed switches.

Software Engineer 2

Sept 2023 - Aug 2024

Citrix | *Led new feature development and backend optimizations for Unified Security.*

- **Optimized API response times by over 95%** (4min to 9s) by re-architecting cloud backend services in C++.
- Drove a **60% increase** in application security license sales by architecting a security recommendation microservice in **Python** and **C++**, deployed on **Docker** and **Kubernetes**, while also serving as the team's **Scrum Master**.
- Saved the threat research team 30+ hours of manual work and **minimized CPU usage by 30%** by leading the **integration of LLMs** (OpenAI) with **Python** and **C** to automate WAF signature extraction and categorization.
- Resolved database bottlenecks with **Splunk**, strategic **indexing** and **refactoring**, enhancing backend efficiency by 66%.
- Mentored and Trained 5 software engineers in Agile, OOP, System Design, and conducted detailed Code Walkthroughs.

Software Engineer 1

Jul 2021 - Sept 2023

Citrix | *Drove innovation by architecting large-scale parallel and distributed systems for application security.*

- **Slashed Software Development Life Cycle runtime by 88%** (16 hrs to under 2 hrs) by architecting a large-scale CI/CD pipeline using Jenkins, **Java**, **Python**, **Groovy** and **Bash Scripting**, earning the **Award** for Technical Innovation.
- Boosted build cycles **30x (1/wk to 30/wk)** with an adaptive scheduling module that parallelized distributed workloads.
- **Eliminated 100%** of false negatives with a context-aware Java module that adaptively selects Ruby and Perl test cases.
- **Accelerated container recovery time by 75%** by using **Prometheus** and **Grafana** for monitoring to trigger **Kubernetes** orchestrated disaster recovery actions.

Software Engineering Intern

Jan 2021 - Jul 2021

Citrix | *Developed new features and built full-stack solutions to accelerate Netscaler's development lifecycle.*

- **Cut kernel test time by 96%** by building a full-stack web app (**Flask**, **React**) for rapid isolated testing.
- Reduced server load by 20% by blocking malicious bots with browser fingerprinting in **JavaScript** and **C++**.
- Slashed processing overhead by 30% while enhancing the positive security model, by optimizing **C-code** within the Netscaler **kernel's core packet engine**, focusing on high-performance **L2/L3** layer packet processing and forwarding.

Technical Skills

- **Languages:** Go, Python, Java, C, C++, C#, HTML/CSS, JavaScript, SQL, Typescript, Ruby, Bash/Shell Scripting
- **Databases:** PostgreSQL, MongoDB (NoSQL), Amazon S3, Pinecone (Vector Databases), Redis, DynamoDB
- **Technologies/Frameworks:** Docker, Kubernetes, Jenkins CI/CD, AWS, Azure, GCP, FastAPI, Flask, Django, ReactJS, Git (Version Control), JIRA, TensorFlow, PyTorch, Spark, Kafka, Elasticsearch, Logstash, Kibana, Terraform, Cursor
- **Skills:** Fullstack Development, Backend Development, System Design, Microservices, REST API, Kernel Development

Education

New York University | *Master of Science in Computer Science* | GPA: 3.95/4.0

May 2026

Manipal Institute of Technology | *Bachelor of Science in Computer Science* | GPA: 3.95/4.0

May 2021